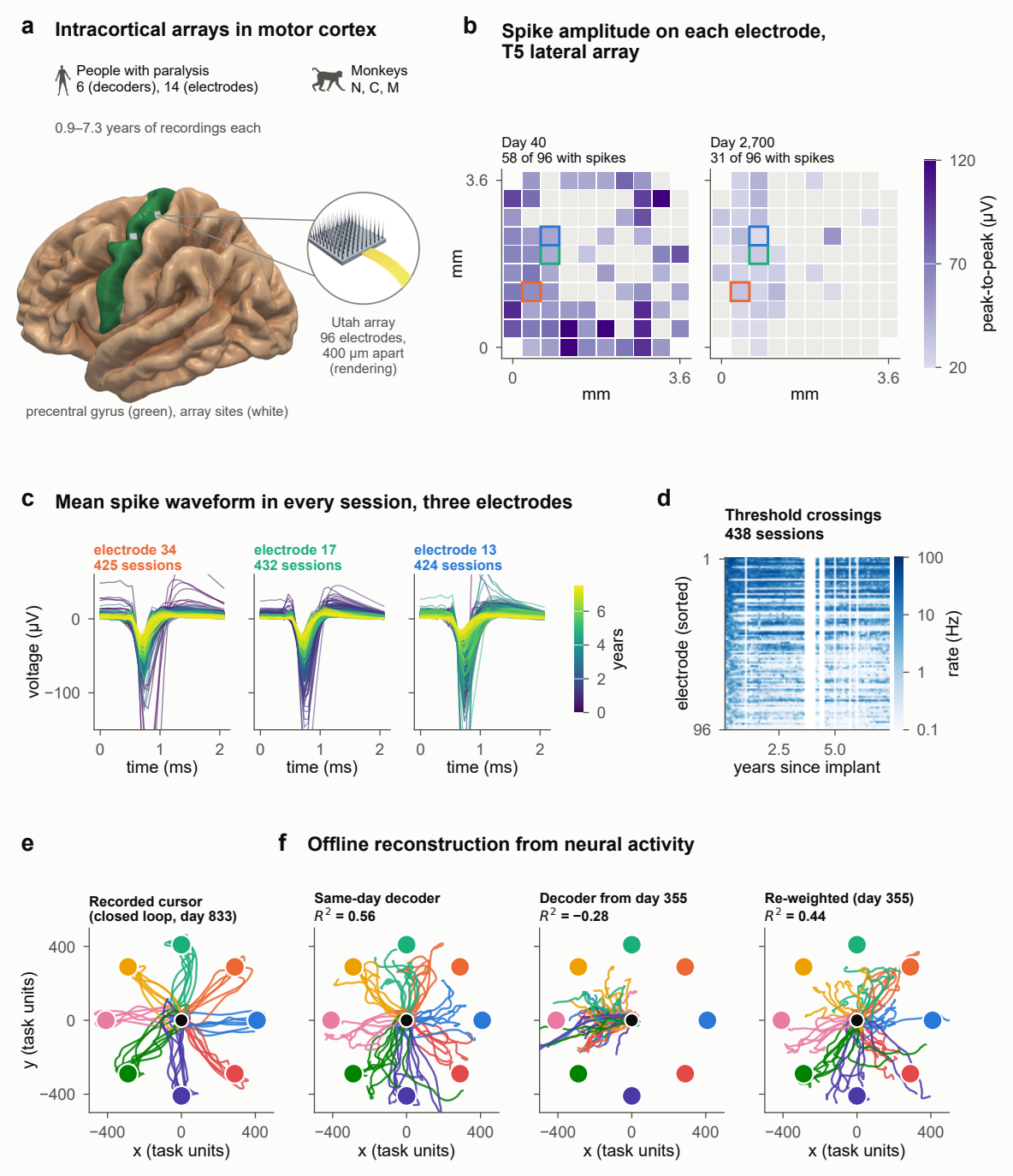

a
Intracortical arrays in motor cortex
People with paralysis
6 (decoders), 14 (electrodes)
Monkeys
N, C, M
0.9–7.3 years of recordings each
Utah array
96 electrodes,
400 µm apart
(rendering)
precentral gyrus (green), array sites (white)
b
Spike amplitude on each electrode,
T5 lateral array
Day 40
58 of 96 with spikes
Day 2,700
31 of 96 with spikes
3.6
0
mm
0
3.6
mm
120
70
20
peak-to-peak (µV)
c
Mean spike waveform in every session, three electrodes
electrode 34
425 sessions
electrode 17
432 sessions
electrode 13
424 sessions
voltage (µV)
0
−100
0
1
2
time (ms)
6
4
2
0
years
d
Threshold crossings
438 sessions
electrode (sorted)
1
96
2.5
5.0
years since implant
100
10
1
0.1
rate (Hz)
e
f
Offline reconstruction from neural activity
Recorded cursor
(closed loop, day 833)
Same-day decoder
R^2 = 0.56
Decoder from day 355
R^2 = −0.28
Re-weighted (day 355)
R^2 = 0.44
y (task units)
400
0
−400
x (task units)